

IML 2026 – Machine Learning Laboratory

LAB 03

Centroid-Based Mean Algorithm and Perceptron

Instructions

1. Complete the lab in Google Colab.
2. Use the Iris dataset.
3. You may use NumPy, Pandas and Matplotlib.
4. You may use `train_test_split()` for splitting the data.
5. You may use `StandardScaler()` for standardization.
6. You may use library functions for accuracy and confusion matrix.
7. The two classification algorithms must be implemented from scratch.
8. Do not use any ready-made classifier for training or prediction.

Dataset

Use the Iris dataset:

<https://archive.ics.uci.edu/dataset/53/iris>

Load the dataset into Google Colab.

Task 1 – Data Preparation

1. Load the dataset.
2. Separate features (X) and target (y).
3. Convert class labels into numerical labels.
4. Split the data into 80% training and 20% testing data.
5. Use `random_state = 42`.
6. Standardize the features using `StandardScaler`.

Task 2 – Centroid-Based Mean Algorithm

Implement the Centroid-Based Mean Algorithm from scratch.

For each class, calculate its centroid:

$$C_k = \frac{1}{N_k} \sum_{i:y_i=k} x_i$$

For each test sample:

1. Calculate its distance from every class centroid.
2. Use Euclidean distance.
3. Assign the sample to the class with the smallest distance.

Report:

- Class centroids
- Test predictions
- Accuracy
- Confusion matrix

Task 3 – Perceptron Algorithm

Implement the Perceptron Algorithm from scratch.

Use:

$$z = w^T x + b$$

and a step activation function for prediction.

Update the parameters using:

$$w = w + \eta(y - \hat{y})x$$

$$b = b + \eta(y - \hat{y})$$

Implement the training procedure yourself.

Since Iris has three classes, implement the multiclass Perceptron using the **One-vs-Rest (OvR)** approach.

Report:

- Number of epochs
- Learning rate
- Training accuracy
- Test accuracy
- Confusion matrix

Task 4 – Comparison

Compare the two algorithms.

Algorithm	Test Accuracy
Centroid-Based Mean	
Perceptron	

Briefly state which algorithm performed better on the test dataset.

Restrictions

Do not use:

- `KNeighborsClassifier`
- `Perceptron`
- Any other ready-made classification algorithm

The following must be implemented by the student:

- Centroid calculation
- Distance calculation
- Centroid-based prediction
- Perceptron prediction
- Perceptron weight and bias updates
- Perceptron training
- Multiclass OvR implementation

Submission

Save the notebook as:

IML2026_Roll-No_Lab_03.ipynb

Upload the completed notebook to Google Classroom before the deadline.